Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/30/2009
PI Name:	Morgan, William F. Ph.D.		
Project Title:	Mitigating High Z Radiation Induced Genomic Instability by Non-Protein Thiols		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Baltimore	**State:**	MD
Zip Code:	21201-1509	**Congressional District:**	7
Comments:	Deceased--PI passed away November 2015		
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	08/12/2005	**End Date:**	07/09/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: PI changed to Janet Baulch in 2008; project continues under Baulch; end date changed to 07/09/2008; see Baulch for FY2009 forward (12/2010) NOTE: End date changed back to 5/31/2009 per S. Krenek/JSC (8/07) End date change to 9/30/2009 per JSC update info (10/06)		
Key Personnel Changes/Previous PI:	PI change to Dr. Janet Baulch, effective 7/10/2008. See Baulch for FY2009 and later reports.		
COI Name (Institution):	Baulch, Janet (University of Maryland)		
Grant/Contract No.:	NNJ06HD31G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	That astronauts will be exposed to the space radiation environment is unavoidable, and all radiation exposure has the potential to elicit a carcinogenic response. Cancer is characterized by gross destabilization of the genome. In vivo and in vitro exposure to radiation can lead to genomic instability. This radiation induced genomic instability shares many of the characteristics associated with genetic instabilities observed in human tumors and may well be process that transforms a normal cell to a neoplastic growth following radiation exposure. We have developed a rapid and sensitive assay for measuring induced genomic instability occurring in the progeny of irradiated cells. One way to protect space travelers from the deleterious effects of radiation exposure is with the use of radioprotective compounds. In this application we will investigate the radioprotective effect of three non-protein thiol drugs already approved for clinical use. This will be achieved by investigating three specific aims: Specific aim 1 will test the hypothesis that non-protein thiols can reduce the frequency of genomic instability induced by High Z particles in human cells. These human cells have been transfected with a sensitive and reliable reporter system to analyze delayed recombination/deletion events associated with radiation exposure in the progeny of surviving cells. Specific aim 2 will test the hypothesis that some of the delayed effects of high Z radiation result from nontargeted bystander effects resulting from soluble factors secreted by irradiated cells. Specific aim 3 will investigate the molecular mechanism(s) by which the non-protein thiols exert their protective effect. Specifically we will test the hypothesis that treating cells with these drugs induces elevated intracellular levels of manganese superoxide dismutase and this reduces iron ion induced genomic instability in RKO36 cells. The goal of these studies is to determine whether drugs already approved for use in the human population can reduce genomic instability induced by radiations commonly encountered in space. If so, then these drugs can potentially be taken by astronauts to protect against deleterious effects associated with space radiations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The basic premise of this application is that astronauts will be exposed to space radiations because there is no way of adequately shielding the spacecraft. Thus, if delayed effects, as manifest as induced genomic instability is a consequence of high LET radiation exposure, and mediated through bystander effects and/or reactive oxygen species, then minimizing the detrimental effects of radiation exposure could be accomplished by the use of effective radical scavengers and antioxidants. While this can be accomplished using classic compounds available in the laboratory situation, e.g., DMSO, BSO etc. there are also a number of drugs used medically in the USA that are effective radical scavengers and/or antioxidants. For example, a number of thiol containing drugs, approved for clinical use e.g., amifostine (WR2721), have well established radioprotective and/or antimutagenic/anticarcinogenic properties. Should we demonstrate that any of these drugs offers statistically significant protection against the detrimental delayed effects of radiation exposure, it is possible that astronauts could take these drugs either prophylactically or immediately after a solar event and decrease the potential health hazards associated with radiation exposure.
Task Progress:	We used WR-1065, the active metabolite of amifostine, to determine how protection from the immediate (direct) effects of high and low-LET radiation exposure influences genomic stability. When added 30 minutes before iron ion irradiation and in high concentration, WR-1065 protected cells from immediate radiation-induced effects as well as from delayed genomic instability. Lower, non-toxic concentration of WR-1065 did not protect cells from death, however it was effective in significantly decreasing genomic instability (delayed effects) in the progeny of irradiated cells. The observed increase in manganese superoxide dismutase protein levels and activity may provide an explanation for this effect. These results confirm that WR-1065 is protective against both low- and high-LET radiation-induced genomic instability in surviving cells. To further investigate the potential molecular mechanisms of this protection, cells were grown to confluence and then exposed to either the radioprotector WR-1065 or the cytokine tumor necrosis factor alpha (TNFa). Total protein was isolated as a function of time up to 32 h after these treatments. Both drugs were effective in elevating intracellular levels of the antioxidant protein SOD2 (also known as MnSOD) as determined by Western blot analysis, while measured SOD2 activity was elevated 5.5 - 6.9-fold. SOD2 reached a maximal level 24 h and 20 h after WR-1065 and TNFa treatments, respectively. Cells were then irradiated in the presence of either WR-1065 or TNFa or were irradiated 24 or 20 h later, respectively, when SOD2 protein levels were highest. The concentrations and exposure conditions used for WR-1065 and TNFa were not cytotoxic and had no effect on plating efficiencies or cell survival compared to untreated controls. The increased survival observed represented a SOD2-mediated delayed radioprotective effect. SOD2 appears to be an important antioxidant gene whose inducible expression is an important element in adaptive cellular responses in general, and the delayed radioprotective effect in particular.
Bibliography Type:	Description: (Last Updated: 02/02/2009)
Articles in Peer-reviewed Journals	Murley JS, Kataoka Y, Baker KL, Diamond AM, Morgan WF, Grdina DJ. "Manganese superoxide dismutase (SOD2)-mediated delayed radioprotection induced by the free thiol form of amifostine and tumor necrosis factor alpha." Radiat Res. 2007 Apr;167(4):465-74. PMID: 17388698 , Apr-2007
Articles in Peer-reviewed Journals	Dziegielewski J, Baulch JE, Goetz W, Coleman MC, Spitz DR, Murley JS, Grdina DJ, Morgan WF. "WR-1065, the active metabolite of amifostine, mitigates radiation-induced delayed genomic instability." Free Radic Biol Med. 2008 Dec 15;45(12):1674-81. PMID: 18845240 , Dec-2008